MATH 202 Your name: _____

Learning target A4, version 1

Use definite integrals to find the mass and center of mass of an object.

Say you have a thin rod that is 6 cm long, and its left end lies at $x = 0$ on the horizontal axis. Suppose that the rod's density is distributed according to the function $\rho(x) = 2e^{-(x-4)^2}$, measured in grams per cm.

1. From the graph of $\rho(x)$ (without doing any computations), do you expect the rod's center of mass, $\bar{x}$, to lie at $x = 3$, left of $x = 3$, or right of $x = 3$? Why?

2. Find the total mass M and center of mass $\bar{x}$ by writing down appropriate definite integrals and evaluating them using technology. Clearly identify your results.

3. How would the rod's mass and center of mass change (if at all) if the density function was instead $\rho(x) = 2e^{-(x-2)^2}$? Don't do any computations, but write a sentence or two to discuss.

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Learning target A4, version 2

Use definite integrals to find the mass and center of mass of an object.

Say you have a thin rod that is 2 feet long, and its left end lies at $x = 0$ on the horizontal axis. Suppose that the rod's density is distributed according to the function $\rho(x) = 5e^{-x^2}$, measured in grams per foot.

(a) From the graph of $\rho(x)$ (without doing any computations), do you expect the rod's center of mass, $\bar{x}$, to lie at $x = 1$, left of $x = 1$, or right of $x = 1$? Why?

(b) Find the total mass M and center of mass $\bar{x}$ by writing down appropriate definite integrals and evaluating them using technology. Clearly identify your results.

(c) Did your work in (b) confirm your intuition in (a)? Why or why not? Write a sentence to discuss.

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Learning target A4, version 3

Use definite integrals to find the mass and center of mass of an object.

The density of cars (in cars per mile) down a 20-mile stretch of the Pennsylvania Turnpike is approximated by

$$\rho(x) = 300\left(2+\sin\left(4\sqrt{x+0.15}\right)\right),$$

where x is measured in miles from the Breezewood toll plaza.

(a) Use Desmos to draw a graph of this function on the interval $[0,20]$. Sketch below.

(b) Find the total number of cars and the "center of mass" by writing down appropriate definite integrals and evaluating them using technology. Clearly identify your results.

(c) What does the "center of mass" mean in the context of cars on a highway?